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GLOBAL CHANGE HYDROLOGY (GCH)

Water Quality in the Water Cycle: Why It Matters

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Water Quality

Definition:

There is no single, universally accepted legal definition of water quality! Several international guidelines and frameworks define it based on suitability for use and protection of human health and ecosystems.

Examples:

1. **World Health Organization (WHO) – Drinking Water Quality:** *"**Water that is safe** for human consumption and meets chemical, microbiological, and radiological standards to protect health."*
2. **European Union – Water Framework Directive (WFD, 2000/60/EC***): "**Good water status** is achieved when water bodies meet ecological and chemical standards that protect ecosystems and human uses."*
3. **US Environmental Protection Agency (EPA) – Clean Water Act:** *"**Water quality** is the physical, chemical, and biological characteristics of water that determine its suitability for designated uses (e.g., drinking, recreation, aquatic life)."*
4. **China – Water Quality Standards:** *"**Multi-layered regulatory framework** for water quality, with standards covering surface water, drinking water, wastewater discharge, groundwater, and marine water."*

Introduction to Water Quality

What is Water Quality and Why Does It Matter?



- **Three pillars of importance:**

1. **Human health** (e.g., safe drinking water, food production).
2. **Ecosystems** (e.g., biodiversity, fisheries, wetlands).
3. **Economy** (e.g., tourism, agriculture, industry).



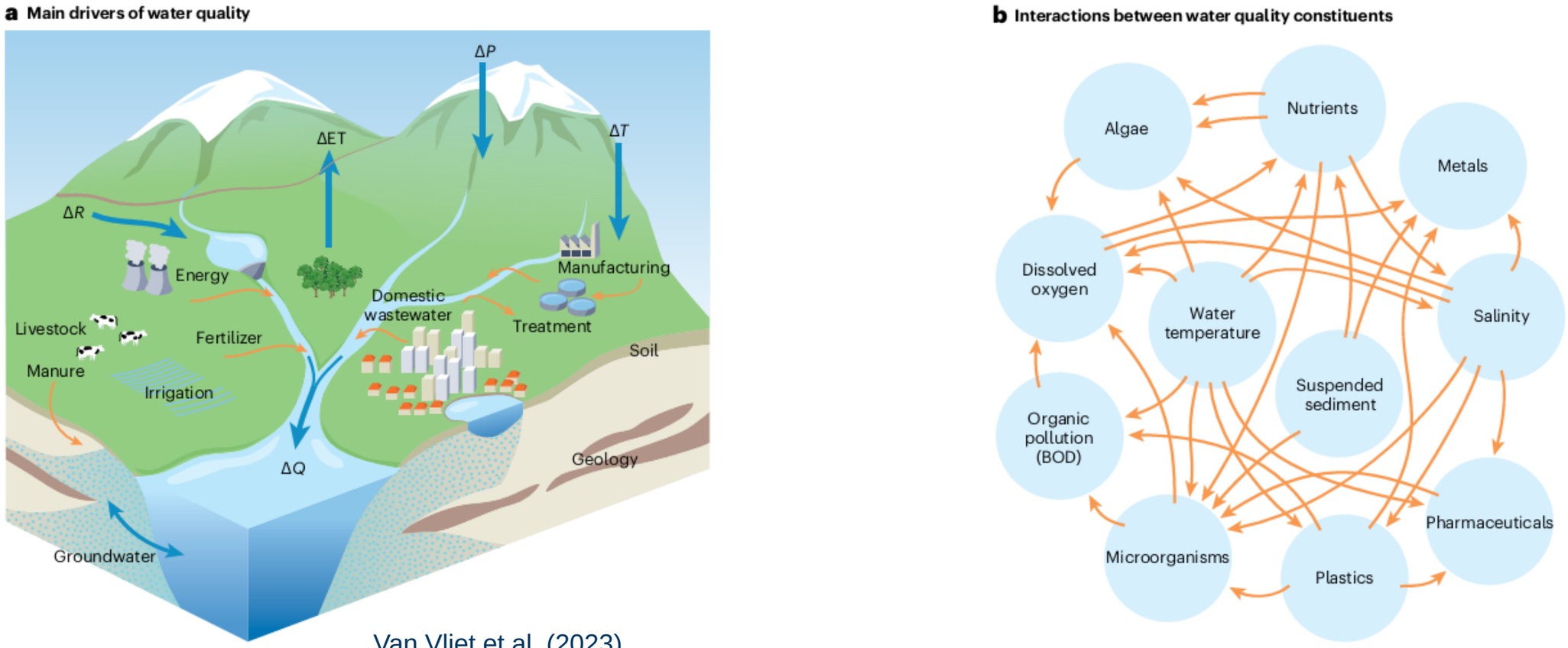




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The Water Cycle and Water Quality

Where Does Water Quality Fit in the Water Cycle?



Van Vliet et al. (2023)

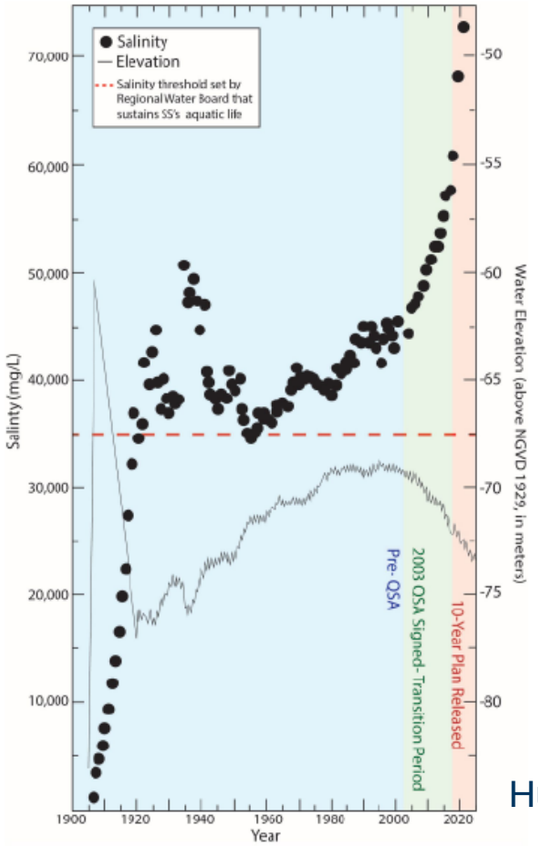
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Water Quantity affects Water Quality

Salton Sea



Source: © NASA



Hung et al. (2024)



Gentle@wikipedia

➢ million fish die in a single day in the Salton Sea (04. August 1999).

Marcum. (1999)

Key Parameters of Water Quality

What Makes Water "Good" or "Bad"?

1. **Physical parameters**
   - Water temperature → affects oxygen levels, aquatic life
   - Turbidity → sediments, algae
2. **Chemical parameters**
   - Heavy metals (lead, mercury) → toxicity
   - PFAS → toxicity
3. **Nutrients**
   - Nitrogen, phosphorus → eutrophication
4. **Biological parameters**
   - Microbial contaminants (E. coli, coliform bacteria)

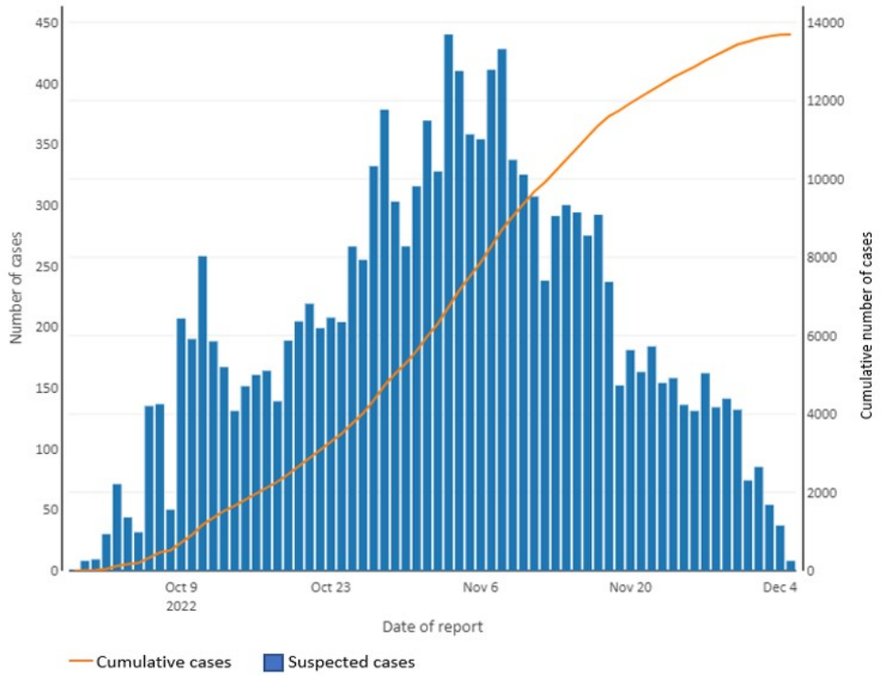
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Why Water Quality Matters for Human Health

The Hidden Threats in Our Water

- **Key points:**

1. **Waterborne diseases**
   - Example: Cholera (Vibrio cholerae)
   - *Case:* 2022 cholera outbreak in Haiti (linked to contaminated water)



www.who.int/emergencies/disease-outbreak-news/item/2022-DON427

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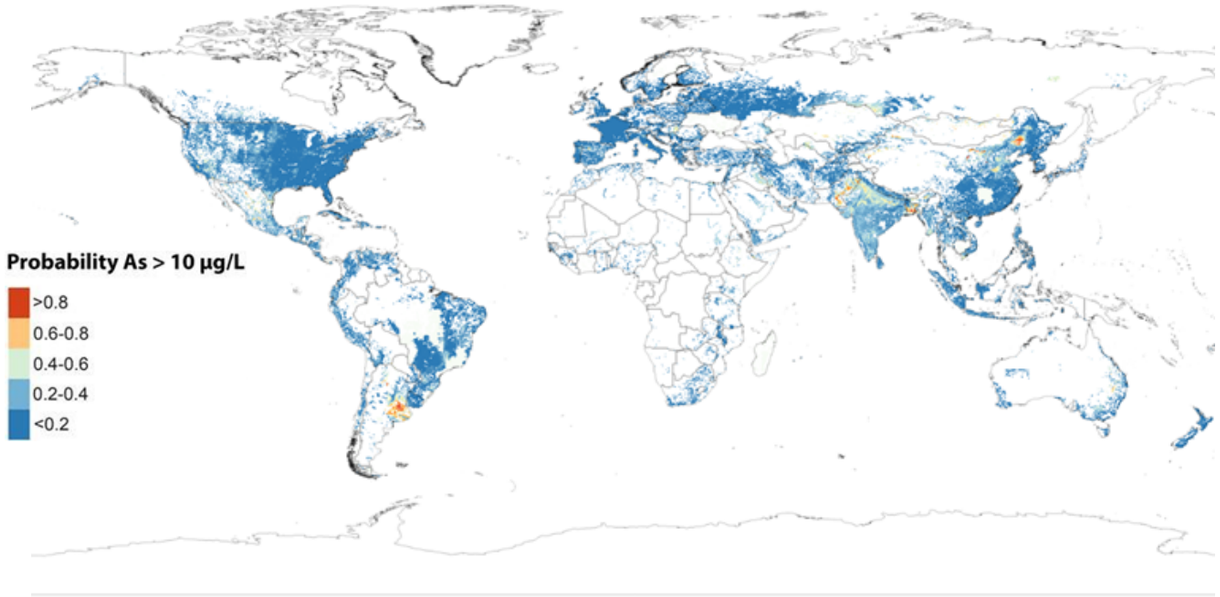
Why Water Quality Matters for Human Health

The Hidden Threats in Our Water

- **Key points:**

2. Chemical contaminants

- Example: Arsenic
- *Case*: natural contamination in Bangladesh/India



- Long-term effects: Cancer, neurological damage, endocrine disruption

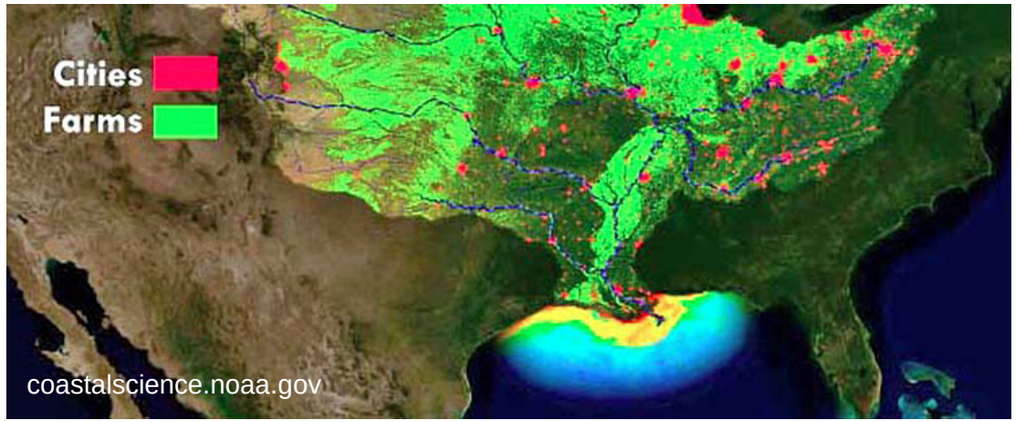
Podgorski & Berg (2020)

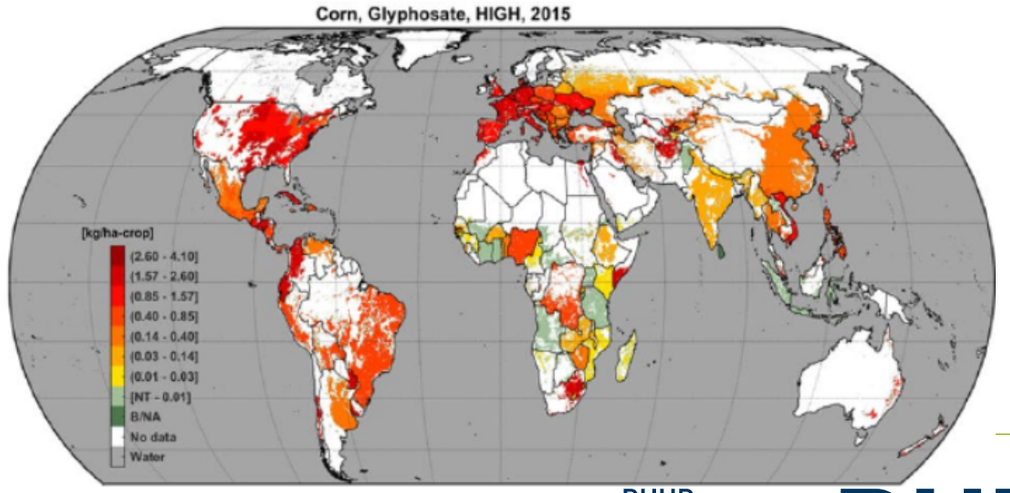
Why Water Quality Matters for Ecosystems

When Water Kills: The Ecological Cost of Pollution

- **Key points:**

1. **Eutrophication**
   - Nutrient overload → algal blooms → oxygen depletion (hypoxia) → "dead zones."
   - *Case:* Gulf of Mexico dead zone (23,000 km²)
2. **Toxic contaminants**
   - Pesticides → neonicotinoids harming bees and aquatic insects
   - *Case*: global application rates of Glyphosate





Maggi et al. (2019)

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Case Study 1 – Drinking Water Contamination

The Flint Water Crisis

April 2014: The Cost-Cutting Switch

Flint, Michigan temporarily switched its municipal water supply to the Flint River to save money. The city failed to add mandatory anti-corrosive agents, ignoring basic water treatment protocols.

The Catalyst: Water Chemistry

The highly corrosive river water rapidly eroded the protective scale lining inside aging municipal pipes. With no barrier left, toxic lead actively leached directly into the residential drinking water supply.

➢ Importance of monitoring, infrastructure maintenance, and public health safeguards

The Human Impact

- ~100,000 residents exposed to highly elevated lead levels.
- Spike in blood lead levels among children, causing irreversible neurodevelopmental damage.
- A fatal outbreak of Legionnaires' disease linked to the water switch.

Systemic Failure & Accountability

Despite months of resident complaints about foul-smelling, discolored water, officials repeatedly denied the problem. In October 2015, the city reconnected to Detroit water. A $626 million legal settlement was later approved for the victims.

Case Study 2 – Eutrophication

Lake Victoria's Algal Blooms: A Regional Crisis



Root Causes & Triggers

- **Anthropogenic Pollution:** Over 70% of basin wastewater enters untreated from major urban hubs like Kampala, Kisumu, and Mwanza.
- **Agricultural Runoff:** Heavy rains wash massive loads of nitrogen and phosphorus from surrounding farms into the catchment area.
- **Climate Catalysts:** Rising temperatures and weakened thermal stratification accelerate the rapid proliferation of toxic blue-green algae.

Ecological & Public Health Toll

- **Aquatic Dead Zones:** Collapsing dissolved oxygen levels trigger widespread fish kills, threatening a critical 300,000-ton annual wild fishery.
- **Cyanotoxin Exposure:** Dominant blooms produce dangerous microcystins, causing severe gastrointestinal, skin, and respiratory issues.
- **Chronic Health Risks:** Recent studies link ingested toxins to rising liver cancer rates; standard rural water treatment plants cannot filter them out.

Case Study 3 – Industrial Pollution

The Kafue River Acid Spill, Zambia (March 2025)

The Cause & Environmental Trigger

- **Industrial Effluent Leak:** A massive release of highly acidic industrial waste bypassed containment protocols in Kitwe, Zambia, entering the Kafue River ecosystem directly.
- **Immediate Eco-Destruction:** The sudden pH drop triggered extensive fish kills along critical stretches of the river and completely destroyed lakeside crops, including vital local maize and groundnut fields.
- **Monitoring Failures:** Delays in detection and the absence of real-time industrial discharge tracking allowed the acidic plume to travel extensively downstream before containment actions began.

Socioeconomic & Public Health Toll

- **Threat to Millions:** Over 60% of the Zambian population relies directly on the Kafue River Basin for domestic water use, intensive agricultural irrigation, and artisanal commercial fishing.
- **Municipal Water Emergency:** The toxic chemical profile forced sudden, absolute shutdowns of municipal water treatment plants to protect public infrastructure from chemical corrosion and prevent human poisoning.
- **Long-Term Vulnerabilities:** Highlights the fragile balance between critical mining/industrial operations and downstream public safety in developing economies, driving demands for stringent legal oversight.

Monitoring and Regulation

How Do We Keep Water Safe?

Who monitors?

- Agencies: EPA (USA), EU-WFD (EU Member States+, WHO (global)
- Local: Water utilities, environmental NGOs

How?

- *Field methods:* Sensors (DO, pH), grab samples, citizen science.
- *Lab analysis:* Spectroscopy, chromatography for contaminants.
- *Remote sensing:* Satellite data for algal blooms/temperature.

Regulations

- Clean Water Act (USA), EU Water Framework Directive, SDG 6 (clean water for all).

Challenges

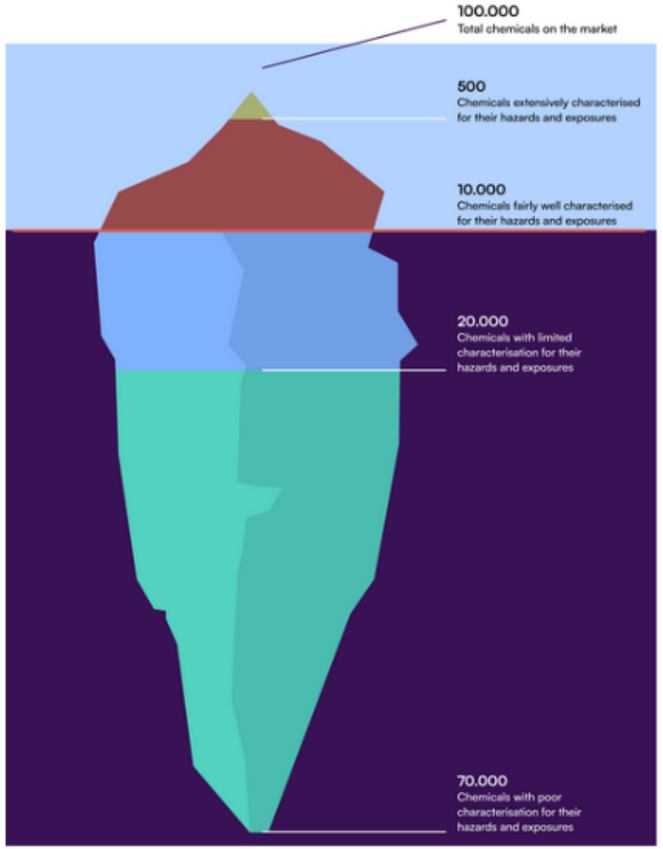
- Limited resources, data gaps, enforcement issues.

Emerging Challenges and Future Directions

New Threats to Water Quality

➢ **Key points:**

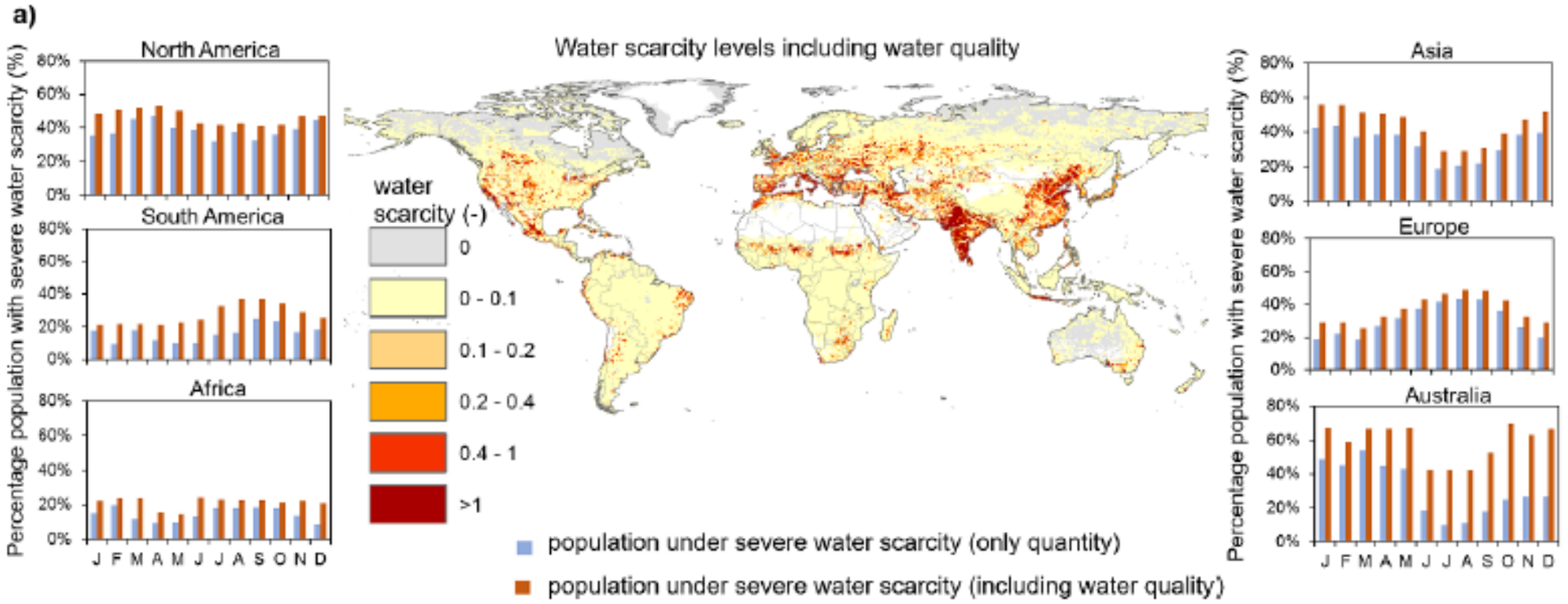
1. **Emerging contaminants**
   - Microplastics (e.g., in bottled water, marine life).
   - Pharmaceuticals (e.g., antibiotics in wastewater → antibiotic resistance).
2. **Climate change impacts**
   - More intense rainfall → increased runoff/pollution.
   - Droughts → higher pollutant concentrations.
   - Warmer water → lower DO, algal blooms



www.eu-parc.eu

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Linking water quantity and water quality



Considering water quantity and water quality in **a water scarcity assessment**, the number of population under severe water stress would increase by **+30%** (annual average)

Water scarcity = f(water quantity, water quality)

- *water temperature, salinity (TDS), organic pollution (BOD), total nitrogen and total phosphorous concentrations*

van Vliet et al. (2021)

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Conclusions

Water Quality: A Shared Responsibility

- **Water quality is dynamic and critical for health, ecosystems, and economies.**
- **Pollution sources: agriculture, industry, urban runoff, climate change.**
- **Solutions: Monitoring, regulation, innovation, community action.**
- **Call to action: "You can be part of the solution!" (e.g., reduce plastic use, support local cleanups).**

